CS1200: Intro. to Algorithms and their Limitations Profs. Anshu and Nesson

Sender–Receiver Exercise 2: Reading for Senders

Harvard SEAS - Fall 2026 *2026-10-19*

1 Solving Graph 2-Coloring using BFS

If you haven't already done so, read the SRE instructions.

The goals of this exercise are:

- to develop your skills at understanding, distilling, and communicating proofs and the conceptual ideas in them, especially for proofs in graph theory
- to reinforce the definition of Graph Coloring and appreciate an application of BFS.

1.1 The Result

Theorem 1.1. *Let G be a graph on n vertices with m edges that can be colored using 2 colors (2-colorable). There is an algorithm to find a 2-coloring of the graph in runtime $O(n+m)$.*

1.2 The Proof

The algorithm is as follows:

```
   BFSColoring(G):
   Input        : A connected graph G = (V, E)
   Output       : A coloring f of G using "few" colors
0  Fix an arbitrary start vertex v_0 ∈ V;
1  Start breadth-first search (BFS) from v_0 to obtain a vertex order v_1, v_2, ..., v_{n-1};
2  foreach i = 0 to n − 1 do
3  |  f(v_i) = min {c ∈ ℕ : c ≠ f(v_j)  ∀j < i s.t. {v_i, v_j} ∈ E}.
4  return f
```

Algorithm 1.1: Greedy coloring in BFS order.

Runtime: The algorithm takes time $O(n+m)$, since BFS takes $o(n+m)$ time and other steps of the algorithm take $O(n)$ time.

Proof of correctness: The intuition behind the proof is simple: a 2-colorable graph is bi-partite and the frontier explored in BFS essentially 'switches' between the two partitions of the graph as the algorithm progresses. Formal details below.

Since G is 2-colorable, fix one such coloring $f^* : V \to \{0,1\}$. Let W be the set of vertices that are assigned the color 0—that is, $W = \{v \in V : f^*(v) = 0\}$—and W' be the set of vertices that are assigned the color 1. Line 3 of the `BFSColoring` algorithm starts with picking up an arbitrary vertex v_0. We may assume that $f^*(v_0) = 0$ without loss of generality, that is $v_0 \in W$. Note that are no edges between the vertices in W and similarly no edges between the vertices in W', by the definition of 2-coloring. (See the example depicted in Figure 1.)

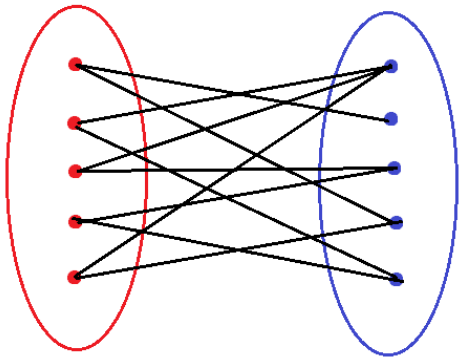


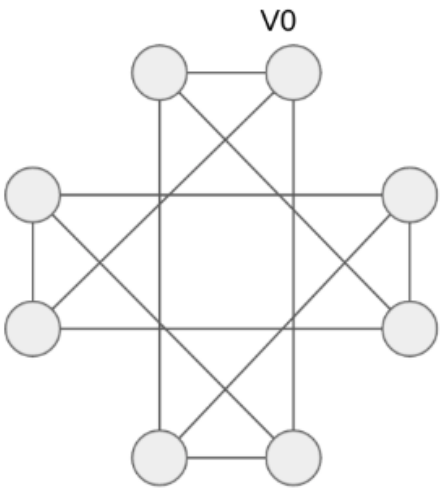
Figure 1: The vertices of a 2-colorable graph can be divided into two parts such that there are no edges within each part. They are also called *bipartite graphs*. As the BFS algorithm is run on such a graph, the frontier vertices switch between the two parts. This is the main idea behind the Theorem

Let f be the coloring returned by the algorithm in Line 4. Let F_d be the set of frontier vertices in d-th iteration of the BFS algorithm (Algorithm 12.1 in HV, Chapter 12). In Line 1, the vertex order from BFS is as follows: $F_0 = \{v_0\}$, followed by all the vertices in F_1, followed by all the vertices in F_2 and so on. We will show the following statement:

"(*)*For even* d, $F_d \subseteq W$ *and for odd* d, $F_d \subseteq W'$. *Furthermore, Line 3 of* `BFSColoring` *ensures that* $f(v) = 0$ *for a vertex* $v \in F_d$ *with even* d *and* $f(v) = 1$ *for a vertex* $v \in F_d$ *with odd* d."

The proof will complete once (*) is established for all d, since the frontier vertices $F_0, F_1, \ldots$ are disjoint and every vertex in v belongs to some frontier vertex in a connected graph. Then (*) will ensure that a proper 2-coloring is achieved, and in fact, that f and f^* are the same coloring.

To prove (*), we use strong induction. Base case with $d = 0$ holds since $F_0 = \{v_0\} \subseteq W$ and $f(v_0) = 0$. Assume that the statement holds for all $d' \leq d$; we will show that it also holds for $d+1$. Suppose d is even. Then for $d+1$, note that every vertex v in F_{d+1} has an edge with some vertex in $F_d \in W$ (this follows from the way the BFS algorithm chooses F_{d+1}). The vertex v must belong to W' (it can't be in W as no two vertices in W share an edge). Thus, $F_{d+1} \subseteq W'$. Finally, we argue that for any vertex $v \in F_{d+1}$, $f(v) = 1$. Recall from the vertex order that the vertices in F_{d+1} are assigned a color in Line 3 after the vertices in $F_0 \cup F_1 \cup \cdots \cup F_d$ are assigned colors. When Line 3 is executed for the vertex v, $f(v)$ is chosen to be the smallest c that is not assigned to any of its neighbors (among the vertices that have been assigned a color so far). However, such neighbors of v only belong to $F_0 \cup F_2 \cup \cdots \cup F_{d-2} \cup F_d$, which are assigned 0 by the induction hypothesis. Thus, $f(v) = 1$ for any vertex $v \in F_{d+1}$. The argument for odd d holds similarly. □



After this proof, run the BFS coloring algorithm from the graph discussed in class - see the figure below. Start from the vertex labelled v_0. How many colors were sufficient?